EA

ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 May/June 2021 Worked Solutions

MayJune 2021 paper rebuilt with the worked solution after each question.

9
paper questions

WMA11
Pure 1

Single-paper solutions
standalone IAL route

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Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 May/June 2021

MayJune 2021 | 9 questions | 75 marks

9
questions

75
marks

Answers
worked solution after each question

Question 1

8 marks

Differentiation

1. The curve C has equation

$$y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5 \qquad x > 0$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$, giving your answer in simplest form.

(4)

The point $P(4, 3)$ lies on C.

(b) Find the equation of the normal to C at the point P. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(4)

(Total 8 marks)

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Worked Solution - Question 1

Differentiation

Topic group

1. Rewrite using powers

$y = \dfrac{x^2}{3} + 4x^{-1/2} + \dfrac{8}{3}x^{-1} - 5.$

2. Differentiate term by term

$\dfrac{dy}{dx} = \dfrac{2x}{3} - 2x^{-3/2} - \dfrac{8}{3}x^{-2} = \dfrac{2x}{3} - \dfrac{2}{x\sqrt{x}} - \dfrac{8}{3x^2}.$

3. Find the tangent gradient at P

At $x = 4$, $\dfrac{dy}{dx} = \dfrac{8}{3} - \dfrac{1}{4} - \dfrac{1}{6} = \dfrac{9}{4}.$

4. Find the normal gradient

The normal gradient is the negative reciprocal of $\dfrac{9}{4}$, so $m_n = -\dfrac{4}{9}.$

5. Write the normal equation

Using $P(4, 3)$, $y - 3 = -\dfrac{4}{9}(x - 4)$. Multiplying by 9 gives $9y - 27 = -4x + 16$, hence $4x + 9y - 43 = 0$.

Final answer

$(a)\ \dfrac{dy}{dx} = \dfrac{2x}{3} - \dfrac{2}{x\sqrt{x}} - \dfrac{8}{3x^2}.$

$(b)\ 4x + 9y - 43 = 0.$

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10 marks

Question 2

Polynomials

2. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

where a is a positive constant.

Given $f(-1) = 32$

(a) (i) show that the only possible value for a is 3

(ii) Using $a = 3$ solve the equation

$$f(x) = 0$$

(5)

(b) Hence find all real solutions of

(i) $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$

(ii) $3\left(9^{3z}\right) + 26\left(9^{2z}\right) - 9\left(9^{z}\right) = 0$

(5)

(Total 10 marks)

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Worked Solution - Question 2

Polynomials

Topic group

1. Use f(-1) = 32

$f(-1) = -a + (6a + 8) + a^2 = a^2 + 5a + 8$. Since $f(-1) = 32$, $a^2 + 5a - 24 = 0$.

2. Choose the positive value of a

$a^2 + 5a - 24 = (a - 3)(a + 8) = 0$. Since a is positive, the only possible value is $a = 3$.

3. Solve f(x) = 0

Using $a = 3$,
$f(x) = 3x^3 + 26x^2 - 9x = x(3x^2 + 26x - 9) = x(3x - 1)(x + 9)$.

4. List the roots

$x = 0$, $x = \frac{1}{3}$, or $x = -9$.

5. Solve the equation in y

Let $u = y^{1/3}$. Then $y = u^3$ and $y^{2/3} = u^2$, so $3y + 26y^{2/3} - 9y^{1/3} = 0$ becomes $3u^3 + 26u^2 - 9u = 0$.

6. Convert back to y

From part (a), $u = -9, 0, \frac{1}{3}$. Therefore $y = u^3$, giving $y = -729, 0, \frac{1}{27}$.

7. Solve the exponential equation

Let $u = 9^z$. Then $3(9^{3z}) + 26(9^{2z}) - 9(9^z) = 0$ becomes $3u^3 + 26u^2 - 9u = 0$.

8. Use positivity of 9^z

Since $9^z > 0$, the only valid root is $u = \frac{1}{3}$. Hence $9^z = \frac{1}{3}$, so $(3^2)^z = 3^{-1}$ and $2z = -1$. Therefore $z = -\frac{1}{2}$.

Final answer

$(a)(i)\ a = 3. \quad (a)(ii)\ x = -9, 0, \frac{1}{3}.$

$(b)(i)\ y = -729, 0, \frac{1}{27}.$

$(b)(ii)\ z = -\frac{1}{2}.$

Question 3

9 marks

Basic Trigonometry

3. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

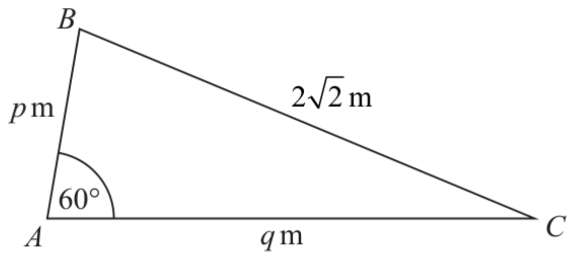

Figure 1

Figure 1 shows the plan view of a flower bed.
The flowerbed is in the shape of a triangle ABC with

- $AB = p$ metres
- $AC = q$ metres
- $BC = 2\sqrt{2}$ metres
- angle $BAC = 60°$

(a) Show that

$$p^2 + q^2 - pq = 8$$

(2)

Given that side AC is 2 metres longer than side AB, use algebra to find

(b) (i) the exact value of p,

(ii) the exact value of q.

(5)

Using the answers to part (b),

(c) calculate the exact area of the flower bed.

(2)

(Total 9 marks)

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Worked Solution - Question 3

Basic Trigonometry

Topic group

1. Apply the cosine rule

$BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos 60°$. Since $BC = 2\sqrt{2}$, $AB = p$ and $AC = q$, this gives $8 = p^2 + q^2 - 2pq\left(\frac{1}{2}\right)$.

2. Show the required result

Therefore $p^2 + q^2 - pq = 8$.

3. Use q = p + 2

Side AC is 2 metres longer than AB, so $q = p + 2$. Substitute into $p^2 + q^2 - pq = 8$.

4. Solve for p

$p^2 + (p+2)^2 - p(p+2) = 8$, so $p^2 + 2p - 4 = 0$. Hence $p = -1 \pm \sqrt{5}$. Since $p > 0$, $p = \sqrt{5} - 1$.

5. Find q

$q = p + 2 = \sqrt{5} + 1$.

6. Find the exact area

Area $= \frac{1}{2}pq\sin 60°$. Now $pq = (\sqrt{5}-1)(\sqrt{5}+1) = 4$, so area $= \frac{1}{2}(4)\left(\frac{\sqrt{3}}{2}\right) = \sqrt{3}\,\mathrm{m}^2$.

Final answer

(a) $p^2 + q^2 - pq = 8.$

(b) $p = \sqrt{5} - 1,\ q = \sqrt{5} + 1.$

(c) $\sqrt{3}\,\mathrm{m}^2.$

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Question 4

6 marks

Integration

4. Find

$$\int \frac{(3\sqrt{x} + 2)(x - 5)}{4\sqrt{x}} \, \mathrm{d}x$$

writing each term in simplest form.

(6)

(Total 6 marks)

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Worked Solution - Question 4

Integration

Topic group

1. Expand the numerator

$(3\sqrt{x}+2)(x-5) = 3x\sqrt{x} - 15\sqrt{x} + 2x - 10.$

2. Divide by 4sqrt(x)

$$\frac{(3\sqrt{x}+2)(x-5)}{4\sqrt{x}} = \frac{3x}{4} - \frac{15}{4} + \frac{1}{2}x^{1/2} - \frac{5}{2}x^{-1/2}.$$

3. Integrate term by term

$$\int\left(\frac{3x}{4} - \frac{15}{4} + \frac{1}{2}x^{1/2} - \frac{5}{2}x^{-1/2}\right)dx = \frac{3x^2}{8} - \frac{15x}{4} + \frac{1}{3}x^{3/2} - 5x^{1/2} + c.$$

4. Write each term simply

So the integral is $\dfrac{3x^2}{8} - \dfrac{15x}{4} + \dfrac{x\sqrt{x}}{3} - 5\sqrt{x} + c.$

Final answer

$$\frac{3x^2}{8} - \frac{15x}{4} + \frac{x\sqrt{x}}{3} - 5\sqrt{x} + c.$$

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Question 5

8 marks

Graphs of Functions

5.

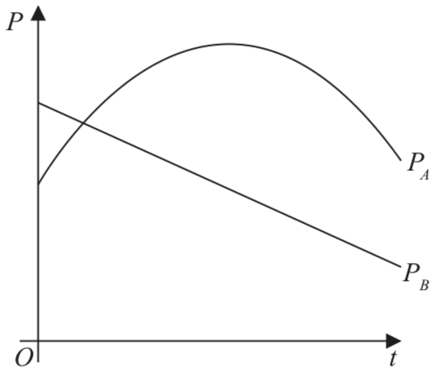

Figure 2

The share value of two companies, company A and company B, has been monitored over a 15-year period.

The share value P_A of **company *A***, in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \qquad t \geqslant 0$$

where t is the number of years after monitoring began.

The share value P_B of **company *B***, in millions of pounds, is modelled by the equation

$$P_B = -1.6t + 44.2 \qquad t \geqslant 0$$

where t is the number of years after monitoring began.

Figure 2 shows a graph of both models.

Use the equations of one or both models to answer parts (a) to (d).

(a) Find the difference between the share value of **company *A*** and the share value of **company *B*** at the point monitoring began.

(2)

(b) State the maximum share value of **company *A*** during the 15-year period.

(1)

(c) Find, using algebra and showing your working, the times during this 15-year period when the share value of **company *A*** was greater than the share value of **company *B***.

(4)

(d) Explain why the model for **company *A*** should not be used to predict its share value when $t = 20$

(1)

(Total 8 marks)

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Worked Solution - Question 5

Graphs of Functions

Topic group

1. Evaluate both models at t = 0

$P_A(0) = 53 - 0.4(0-8)^2 = 53 - 25.6 = 27.4$. Also $P_B(0) = -1.6(0) + 44.2 = 44.2$.

2. Find the initial difference

The difference is $44.2 - 27.4 = 16.8$, so the share values differ by 16.8 million pounds.

3. State the maximum of company A

$P_A = 53 - 0.4(t-8)^2$ has maximum value 53, reached when $t = 8$, which is within the 15-year period.

4. Set up the inequality for company A greater than company B

$53 - 0.4(t-8)^2 > -1.6t + 44.2$.

5. Rearrange to a quadratic inequality

Expanding gives $-0.4t^2 + 8t - 16.8 > 0$. Multiplying by -2.5 reverses the inequality: $t^2 - 20t + 42 < 0$.

6. Find the crossing times

$t = \dfrac{20 \pm \sqrt{400-168}}{2} = 10 \pm \sqrt{58}$. Therefore company A is greater between the two roots.

7. Restrict to the 15-year period

Since the model is monitored over $0 \le t \le 15$, the required times are $10-
\sqrt{58}

8. Explain the limitation

The model should not be used at $t = 20$ because it was only built from a 15-year monitoring period; 20 years is outside that range.

Final answer

(a) 16.8 million pounds.

(b) 53 million pounds.

$(c)\ 10-\sqrt{58}

(d) $t = 20$ is outside the 15-year model.

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8 marks

Question 6

Integration

6. The curve C has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $\mathrm{f}'(x) = \dfrac{32}{3x^2} + 3 - 2\left(\sqrt[3]{x}\right)$

(a) find the equation of the tangent to C at P. Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

(b) Find, in simplest form, $\mathrm{f}(x)$.

(5)

(Total 8 marks)

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Worked Solution - Question 6

Integration

Topic group

1. Find the gradient at P

$f'(x) = \dfrac{32}{3x^2} + 3 - 2\sqrt[3]{x}$. At $x = 8$,

$f'(8) = \dfrac{32}{3(64)} + 3 - 2(2) = \dfrac{1}{6} - 1 = -\dfrac{5}{6}$.

2. Write the tangent equation

Using $P(8,2)$, $y - 2 = -\dfrac{5}{6}(x-8)$, hence $y = -\dfrac{5}{6}x + \dfrac{26}{3}$.

3. Integrate f'(x)

$f'(x) = \dfrac{32}{3}x^{-2} + 3 - 2x^{1/3}$, so $f(x) = -\dfrac{32}{3x} + 3x - \dfrac{3}{2}x^{4/3} + c$.

4. Use P(8,2) to find c

$2 = -\dfrac{32}{24} + 24 - \dfrac{3}{2}(16) + c = -\dfrac{4}{3} + c$, so $c = \dfrac{10}{3}$.

5. Write f(x)

$f(x) = -\dfrac{32}{3x} + 3x - \dfrac{3}{2}x^{4/3} + \dfrac{10}{3}$.

Final answer

$(a)\ y = -\dfrac{5}{6}x + \dfrac{26}{3}$.

$(b)\ f(x) = -\dfrac{32}{3x} + 3x - \dfrac{3}{2}x^{4/3} + \dfrac{10}{3}$.

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Question 7

10 marks

Radians

7.

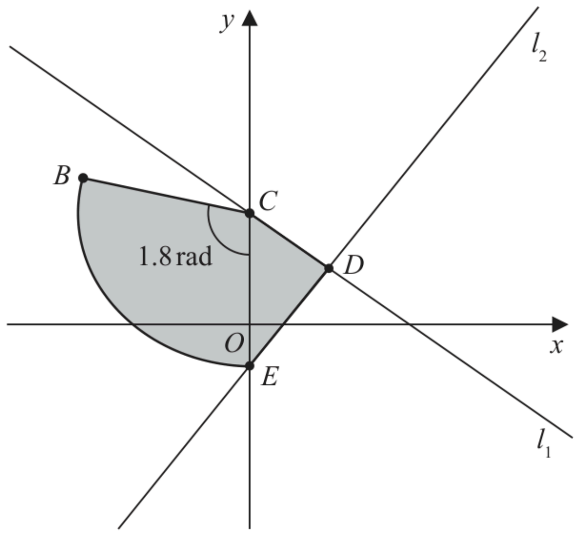

Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y-axis at the point C, as shown in Figure 3.

(a) State the y coordinate of C.

(1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y-axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE.

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE.

(d) Calculate the exact area of the region $CBED$.

(3)

(Total 10 marks)

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Worked Solution - Question 7

Radians

Topic group

1. Find C

On the y-axis, $x = 0$. From $4y + 3x = 48$, $4y = 48$, so the y-coordinate of C is 12.

2. Find the gradient of l1

$4y + 3x = 48$ gives $y = -\frac{3}{4}x + 12$, so the gradient of l_1 is $-\frac{3}{4}$.

3. Use perpendicular gradients for l2

Since l_2 is perpendicular to l_1, its gradient is $\frac{4}{3}$. Through $D(8, 6)$, $y - 6 = \frac{4}{3}(x - 8)$.

4. Show the y-coordinate of E

$y = \frac{4}{3}x - \frac{32}{3} + 6 = \frac{4}{3}x - \frac{14}{3}$. Therefore when $x = 0$, $E_y = -\frac{14}{3}$.

5. Find the radius of the sector

$C = (0, 12)$ and $E = \left(0, -\frac{14}{3}\right)$, so $CE = 12 + \frac{14}{3} = \frac{50}{3}$.

6. Find arc BE

For sector BCE, $r = \frac{50}{3}$ and $\theta = 1.8 = \frac{9}{5}$. Hence arc $BE = r\theta = \frac{50}{3} \cdot \frac{9}{5} = 30$.

7. Find the sector area

Sector area $= \frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{50}{3}\right)^2\left(\frac{9}{5}\right) = 250$.

8. Find the triangle area and total area

Triangle CDE has vertical base $CE = \frac{50}{3}$ and horizontal height 8, so its area is $\frac{1}{2} \cdot \frac{50}{3} \cdot 8 = \frac{200}{3}$. Total area $= 250 + \frac{200}{3} = \frac{950}{3}$.

Final answer

(a) 12.

(b) $E_y = -\frac{14}{3}$.

(c) 30.

(d) $\frac{950}{3}$.

Question 8

9 marks

Polynomials

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

(a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a, b and c are constants to be found.

(3)

The point P is the minimum point of C_1

(b) Deduce the coordinates of P.

(1)

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A, B, C and D are constants.

Given that C_2

- passes through P
- intersects the x-axis at -4, -2 and 3

(c) find, making your method clear, the values of A, B, C and D.

(5)

(Total 9 marks)

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Worked Solution - Question 8

Polynomials

Topic group

1. Complete the square

$3x^2 + 6x + 9 = 3(x^2 + 2x) + 9 = 3\big((x+1)^2 - 1\big) + 9 = 3(x+1)^2 + 6.$

2. Deduce the minimum point

The minimum occurs when $(x+1)^2 = 0$, so $x = -1$ and $y = 6$. Therefore $P = (-1, 6)$.

3. Use the x-intercepts of C2

Since C_2 intersects the x-axis at $-4, -2$ and 3, it has form $y = k(x+4)(x+2)(x-3)$.

4. Use point P to find k

Substitute $P = (-1, 6)$: $6 = k(3)(1)(-4) = -12k$, so $k = -\dfrac{1}{2}$.

5. Expand the cubic

$(x+4)(x+2)(x-3) = (x^2 + 6x + 8)(x-3) = x^3 + 3x^2 - 10x - 24.$

6. Read off A, B, C and D

Thus $y = -\dfrac{1}{2}x^3 - \dfrac{3}{2}x^2 + 5x + 12$, so $A = -\dfrac{1}{2}$, $B = -\dfrac{3}{2}$, $C = 5$ and $D = 12$
.

Final answer

$(a)\ 3(x+1)^2 + 6.$

$(b)\ P = (-1, 6).$

$(c)\ A = -\dfrac{1}{2},\ B = -\dfrac{3}{2},\ C = 5,\ D = 12.$

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7 marks

Question 9

Trigonometric Functions

9.

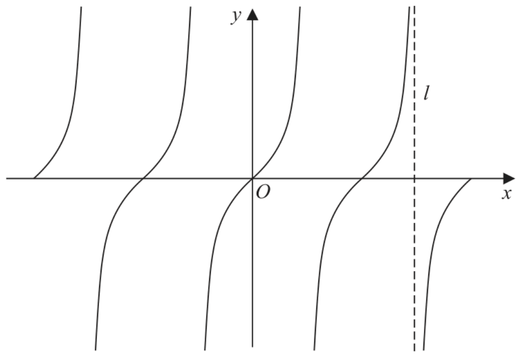

Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \qquad -2\pi \leqslant x \leqslant 2\pi$$

The line l, shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l.

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \qquad -2\pi \leqslant x \leqslant 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leqslant x \leqslant 2\pi$

(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leqslant x \leqslant 40\pi$

(ii) $-10\pi \leqslant x \leqslant \frac{5}{2}\pi$

(2)

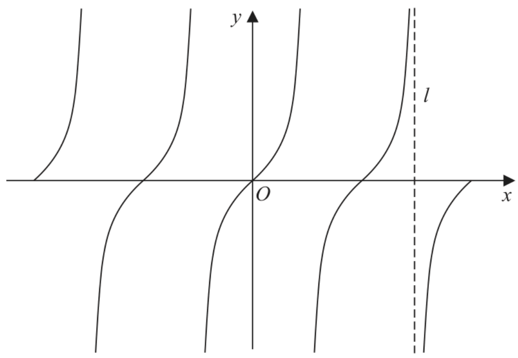

Diagram 1

(Total 7 marks)

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Worked Solution - Question 9

Trigonometric Functions

Topic group

1. State the shown tangent asymptote

The shown dashed asymptote is the right-hand positive asymptote before 2π, so l has equation $x = \dfrac{3\pi}{2}$.

2. Describe the reciprocal graph

For $y = \dfrac{1}{x} + 1$, the vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 1$. The right branch lies above $y = 1$ and the left branch lies below $y = 1$.

3. Count intersections on -2pi to 2pi

The number of solutions of $\tan x = \dfrac{1}{x} + 1$ is the number of intersections of the two graphs. From the sketch on $-2\pi \leq x \leq 2\pi$, there are 5 intersections.

4. Count solutions for 0 <= x < 40pi

From the repeated tangent branches and the reciprocal curve, there is one intersection in each positive interval between consecutive tangent asymptotes. From 0 to 40π, this gives 40 solutions.

5. Count solutions for -10pi <= x <= 5pi/2

Extending the same intersection pattern across the stated interval gives 14 intersections, so there are 14 solutions.

Final answer

(a) $x = \dfrac{3\pi}{2}$.

$(b)(i)$ $y = 1$ is the horizontal asymptote.

$(b)(ii)$ 5.

$(c)(i)$ 40.

$(c)(ii)$ 14.